

An Analysis of Trade Credit Behaviour of Indian Firms

South Asia Economic Journal

22(1) 132–154, 2021

© 2021 Research and Information System
for Developing Countries & Institute of

Policy Studies of Sri Lanka

Reprints and permissions:

in.sagepub.com/journals-permissions-india

DOI: 10.1177/13915614211009659

journals.sagepub.com/home/sae

Nitin Kumar¹, Arvind Shrivastava², Purnendu Kumar² and M. Ishaq Bhatti³

Abstract

Trade credit transactions are quite common for businesses. The article carries out the trade credit analysis for an emerging economy, namely Indian corporate sector employing rich information dataset covering multiple industries such as manufacturing, services, construction and others, since the period of financial crisis including both firm specific and macro-economic factors. The annual dataset spans 13 years from 2006 to 2018 covering the crisis period. Applying dynamic panel framework, it is found that the inventory management and macro indicators are significant in determining trade credit for Indian firms. While trade payable is chiefly driven by raw material inventory, firms having reasonable stock of raw or finished goods inventory are less likely to offer trade credit. Large-sized firms are found to be both leading consumers and suppliers of the trade credit. The pecking order theory is clearly validated with net profits being preferred over the trade credit that is a more expensive source of finance. Credit from formal financial sources is found to act as a substitute to trade credit borrowing.

JEL: G3, G21, E4, C23

Keywords

Trade credit, corporate finance, bank borrowing, macro economy, panel data, generalized method of moments

¹ Reserve Bank of India, Lucknow, Uttar Pradesh, India.

² Reserve Bank of India, Mumbai, Maharashtra, India.

³ College of Arts, Social Sciences and Commerce, Latrobe University, Australia.

Corresponding author:

Nitin Kumar, Reserve Bank of India, Lucknow, 8–9 Vipin Khand, Gomtinagar, Lucknow 226010, Uttar Pradesh, India.

E-mail: nitin_005us@yahoo.com

Trade payables and receivables represent the sources and uses of funds of a firm. Together these are also known as trade credit and are critical financing instruments of companies. The distinctive feature of the trade credit lies in its convenience. It is not always feasible for a business concern to obtain external funding from formal credit institutions as it involves factors such as collateral, reputation, income generation capacity and processing time. The trade credit is an agreement between buyer and seller, which allows buyer to delay payment for the goods and/or service supplied. The trade credit is a short-term cash management tool wherein a firm can act as both supplier and user of the trade credit simultaneously. Trade credit receivables depict the amount lent by a firm to other firms. Whereas trade credit payable are the borrowings of a company from its suppliers. It is often observed that firms receive the trade credit from their own suppliers are more likely to extend trade credit to their customers. Literature supports the view that suppliers lend more liberally to borrowers especially during downturns due to availability of superior information about credit worthiness of borrowers overcoming moral hazard problem of lending (Biais & Golliers, 1997; Burkart & Ellingsen, 2004; Smith, 1987). However, the trade credit is not sans its limitations. It is a short-term liquidity management tool and the expensive form of the credit beyond its discount period (Klapper et al., 2011; Petersen & Rajan, 1997). Firms that are credit rationed or constrained in their accessibility to formal sources of finance are more likely to utilize trade credit indicating weak financial institutions (Casey & O'Toole, 2014). Excessive transaction of the trade credit is also believed to be related with contractionary economic activity (Coulibaly et al., 2013; Love & Zaidi, 2010).

In inefficient and less developed financial systems exhibiting tightened credit and monetary policies, firms may have to seek alternative sources of funds. Despite being costly post discount period, the trade credit is the vital option of finance as documented by various authors. Employing global firm level datasets for later part of the 1980s, Petersen and Rajan (1997) reported that accounts receivable to sales ratio for small firms in the USA stood at 7.3% versus 18.5% for larger firms. Likewise, there existed predominance of the trade credit among bigger firms, with accounts payables to sales ratio being only 4.4% for small firms compared to 11.6% for larger firms. For firms operating in UK's manufacturing sector over 1993–2003, Bougheas et al. (2009) reported account receivables to sales of 17%, whereas account payables to sales stood at 10%. Using Euro area firm-level data Casey and O'Toole (2014) found the highest usage of the trade credit in Ireland, a country which has simultaneously experienced a severe banking crisis and sovereign debt funding crisis with 75% of firms opting for the alternative finance. The second highest usage is in Finland at 66% followed by Greece at 61%. The country with lowest reported usage of the trade credit is France at just 15% of firms utilizing it. As per Ghosh (2015), the trade credit usage as percentage of total funding was roughly 16% for Indian manufacturing firms during 1993–2012.

In this context, the study endeavours to understand the trade credit characteristics of Indian companies. It undertakes its analysis across various industries, ownership groups and so on. Employing panel dataset for the period 2006–2018, it ascertains the existence of structural changes as regards to the trade credit for Indian industries, the role of bank borrowings vis-a-vis other sources of funds for Indian firms. It adds to the literature by including macro-economic characteristics of the economy, which enable us to understand the relationship of the trade credit with business cycles. It is revealed that lately foreign owned firms have the highest operations of the trade credit followed by the joint sector. Analytical results establish the existence of the long-term trade credit relationship among peers. Procurement of raw material is mainly on the trade credit loan whereas companies stocking reasonable level of finished goods are less likely to offer the trade credit for selling their product. Enterprises enjoying higher liquidity are both generous user and supplier of the trade credit. Moreover, the macro-economic parameters do seem to drive the trade credit usage decisions for certain sectors. In the next section, we discuss the prominent research studies exploring the subject from various perspectives. Exposition of data and empirical methodology are provided in subsequent sections, respectively. Discussion of results is presented in the following section. The overall findings and policy implications are summarized in the last section.

Review of Literature

There exists considerable work globally on the trade credit that posits varied motives for the existence of the trade credit despite specialized institutions for the purpose. As per one strand of the literature, the trade credit is a source of information/signalling about the credit worthiness and market power of a typical firm (Biais & Gollier, 1997; Petersen & Rajan, 1997). As per other school of thought, it is an alternative source of fund, where either the financial markets are not well developed or firms are constrained in terms of their accessibility to the formal financial system for certain reasons (Petersen & Rajan, 1997; Wilner, 2000). Researchers such as Burkart and Ellingsen (2004), Wilner (2000) have extensively analyzed the trade credit as substitute/complement to bank and debt financing. The importance of the trade credit in influencing performance has been carried out by Du et al. (2008), Kumar & Shrivastava (2013). Still others have documented the role of the trade credit in leading to financial crisis (Love & Zaidi, 2010; Molina & Preve, 2009).

In a game theoretical setup with three agents, namely the bank, seller and buyer, Biais and Gollier (1997) solved for the Bayesian Nash equilibrium and showed that trade credit enables the private information of the seller to be used in the lending relationship with banks. They also proposed that the trade credit can transmit a seller's private information to banks and can alleviate the bank credit rationing due to the asymmetric information between banks and firms.

Examining the trade credit for US firms based on National Survey of Small Business Finances for 1988–1989, Petersen and Rajan (1997) found that small firms which do not have long-term banking relationships, resort more on the trade credit as a significant source of financing. They suggested that suppliers lent to constrained firms because they have a comparative advantage in getting information about buyers, they can liquidate assets more efficiently and have an implicit equity stake in the firms. Also, results included that firms with better access to the credit offer more the trade credit.

Wilner (2000) has the framed model for lending and renegotiation strategies for firms. It suggests that the trade credit can serve as a substitute to the bank credit. Burkart and Ellingsen (2004) argued that bank and trade credit may be compliments or substitutes depending on the borrower's wealth, which is likely to be correlated with size. Firms with wealth below a certain critical level exhaust both their bank and trade credit and invest less than the optimal level. For these firms, the trade credit availability follows the bank credit. They also showed that profitable, but finance constrained firms would prefer not to offer the trade credit.

Du et al. (2008) used World Bank data of Chinese firms and investigated the relative importance of bank loans and trade credit in promoting the firm performance. They found that access to bank loans is central to improving firm performance and growth, while availability of the trade credit is much less important.

Molina and Preve (2009) studied the use of supplier's trade credit by firms in financial distress and found firms in financial distress use a significantly larger amount of the trade credit to substitute alternative sources of financing. They also found that the firms which are smaller, with less market power, and with more unique products tend to use more trade credit financing when in distress. They further studied the trade receivables policy of distressed firms as the trade-off between firm's willingness to gain sales and the firm's need for cash. They found that firms increase trade receivables when they have profitability problems, but reduce trade receivables when they have cash flow problems.

Love and Zaidi (2010) examined whether suppliers extend more credit to constrained customers during crisis, or whether they withdraw the credit from constrained firms due to their high-credit risk utilizing about 3000 firms from four East Asian countries during 1998 financial crisis. The results do not support hypothesis that the trade credit is viable substitute for contracted bank credit in times of the financial crisis. Also, it is indicated that the trade credit usage becomes more expensive and restrictive during crisis.

The trade credit analysis literature for Indian conditions is quite limited. Vaidya (2011) analyzed trade credit determinants for Indian manufacturing firms as obtained Prowess database from 1993 to 2006. The strong evidence was established in support of an inventory management for the existence of the trade credit with highly profitable firms found to both extend and utilize less trade credit. Trade credit in general was seen as financial response to variable demand for its finished goods. The study utilized generalized method of moments (GMM) to control for possible endogeneity of regressors. It did not account for the dynamic nature of the response variable as has been performed in the present

exposition. Moreover, the present study employs richer information set with more type of firm classifications.

Another study that has studied the association between the trade credit and bank credit is carried by Ghosh (2015). Focusing on the Indian manufacturing sector during 1993–2012 the author finds the complementary relationship between trade and bank credit. However, the final decision of a firm regarding the trade credit depends on its degree of financial constraints faced during the business cycles.

The trade credit has been examined from a different perspective for Indian private companies also. Kumar and Shrivastava (2013) found evidence in support of influence of the trade credit on profitability for Indian firms. While trade receivables had a positive influence of company profitability, an inverse relation was obtained between trade payables and performance for small-sized businesses.

Data and Key Variables

The study is based on annual information of non-government and non-financial firms extracted from Centre for Monitoring Indian Economy Prowess database, which is compiled based on annual reports of companies. The government and financial companies including banks are excluded from the analysis due their separate regulatory and operational structure. Additionally, only those entities are retained that have at least six years of existence and three years of continuous data availability for consistency purpose. The unbalanced panel includes 8975 companies for the period 2006–2018 comprising a total of 86,227 observations.

As our key focus is the trade credit, so our dependent variables are trade payable (TP) and trade receivable (TR) representing trade credit liabilities and assets, respectively. Both the variables are normalized by total assets. Other explanatory indicators that are examined are described as follows.

Size of the firm is captured by natural logarithm of real sales¹ denoted as SIZE. Firm size is a proxy for creditworthiness and reputation of a firm (Petersen & Rajan, 1997). Typically, larger firms borrow more even though they have higher cash flows and fewer growth opportunities. This suggests that they enjoy more creditworthiness. So, SIZE is anticipated to have positive impact both on AP and AR.

Inventory is the critical parameter that may drive the trade credit in either direction. Inventory can act as collateral to obtain the trade credit and has been found to have a positive relation with accounts payables by Cuna (2007). But Bougheas et al. (2009) found inventories having the negative effect on account receivables as higher stocks act as incentive to increase sales by net contributor of the trade credit allowing buyers to delay the payment. As suggested by Vaidya (2011) finished goods inventory may have greater role for determining trade receivable, whereas trade payable may be more sensitive to raw material inventory. So, we include two variables for inventory, namely raw materials inventory (RAW_INV) and finished goods inventory (FIN_INV) to study their impact separately. Both the variables are scaled by assets.

Impact of profitability is measured by return on assets (ROA).² The literature provides mixed results for profits with Petersen and Rajan (1997) reporting negative relation between accounts receivables and net profits. However, Bougheas et al. (2009) found the positive impact of profits for both account receivables and account payables.

There exists mixed evidence about the relationship between bank loan and trade credit. Petersen and Rajan (1997) establish that the trade credit is more expensive compared to bank borrowing, thereby occupying lower position in pecking order of finance for the US scenario. Burkart and Ellingsen (2004) suggested that association between the two can be complementary for firms with constrained debt capacity and substitute for units with sufficient aggregate debt capacity. The bank credit acts as a substitute with account receivable, whereas it shares a complementary relationship with account payable in the pre-crisis period for Indian manufacturing enterprises (Vaidya, 2011). To ascertain the impact of the bank credit, loan from banks (BK_LOAN) is included, which is defined as bank borrowings to assets of a firm.

Apart from banks there exist other specialized institutions and mediums also for corporate financial needs. To comprehend the association of the trade credit with other forms of credit, we have included other loans to asset (OTH_LOAN). The loans from other financial institutions, amount raised through issue of bonds and debentures, external borrowings and so on, are the major components included in other loans.

Current assets to total assets (CATA) have been included to measure liquidity. It is an indicator that measures firm's ability to repay its debts in near future. Higher ratio implies the comfortable liquidity position and less dependence on the external financing.

Burkart and Ellingsen (2004) posited that the trade credit has relationship with economy business cycles. A number of macroeconomic factors believed to affect credit environment and alternative finance were chosen by Casey and O'Toole (2014) such as GDP growth, the ratio of private sector credit to GDP, yield on 10-year sovereign bond, credit default swap rate. Yang (2011) reported that during periods of monetary tightening, the trade credit can substitute for the bank credit, whereas the two factors act as complementary during periods of monetary easing for US manufacturing firms. In this context, the GDP growth rate (GR_RATE), consumer price index of industrial workers³ (INF) and weighted average call money rate (INT_RATE) are taken as macroeconomic indicators for the economic growth, inflation and lending rate, respectively. The interest rate is a key variable for firms' financial decisions.

Methodology

In the present pooled information set, the firm level characteristics may not be independent over time rendering the ordinary least-square technique inappropriate, which may provide biased and inconsistent estimates. Additionally, firms may share the trade credit relationship among peer that may extend over period due to

factors such as trustworthiness and confidence (Casey & O'Toole, 2014). So, a dynamic framework is employed wherein the trade credit of the previous period is also taken as an independent variable. In dynamic panel models with large number of cross-sections and small time period, the GMM estimator developed by Arellano and Bond (1991) and Arellano and Bover (1995) are known to provide efficient estimates. Bougheas et al. (2009) and Vaidya (2011) have also applied the GMM estimation strategy for the trade credit analysis. The GMM is formulated as follows.

$$y_{i,t} = \alpha y_{i,t-1} + \beta' X_{i,t} + K_i + \varepsilon_{i,t}; \text{ where } \varepsilon_{i,t} \sim N(0, \tau^{-1}) \text{ and } \tau = \frac{1}{\sigma^2} \quad (1)$$

In Equation (1), $y_{i,t}$ depicts the dependent variable of i th firm at t th period. The lagged endogenous variable is represented by $y_{i,t-1}$. The matrix $X_{i,t}$ depicts the values of explanatory variables. K_i is the unobserved firm effect and $\varepsilon_{i,t}$ being the usual idiosyncratic shocks. All specifications are estimated using GMM first-difference specification, so that firm specific effects are removed. So, Equation (1) can be re-formulated as follows:

$$y_{i,t} - y_{i,t-1} = \alpha (y_{i,t-1} - y_{i,t-2}) + \beta' (X_{i,t} - X_{i,t-1}) + (\varepsilon_{i,t} - \varepsilon_{i,t-1}).$$

Under the assumptions of: (a) error term is not serially correlated, (b) explanatory variables are not correlated with future realizations of the error term, the GMM dynamic panel estimator employs the following moment conditions.

$$E[y_{i,t-s} \cdot (\varepsilon_{i,t} - \varepsilon_{i,t-1})] = 0 \quad \text{for } s \geq 2; t = 3, \dots, T.$$

$$E[X_{i,t-s} \cdot (\varepsilon_{i,t} - \varepsilon_{i,t-1})] = 0 \quad \text{for } s \geq 2; t = 3, \dots, T.$$

The application of instruments in the GMM difference estimation enables to manage not only the issue of endogeneity but also the correlation of $(y_{i,t-1} - y_{i,t-2})$ with $(\varepsilon_{i,t} - \varepsilon_{i,t-1})$. Additionally, difference GMM also circumvents the issue of number of instruments, which may be otherwise large in case of the system GMM leading the Sargan test to be weak.

The detailed specifications of Equation (1) are as follows:

$$\begin{aligned} TP_{i,t} = & \alpha TP_{i,t-1} + \beta_1 SIZE_{i,t} + \beta_2 RAW_{INV i,t} + \beta_3 FIN_{INV i,t} \\ & + \beta_4 ROA_{i,t} + \beta_5 OTH_{LOAN i,t} + \beta_6 BK_{LOAN i,t} \\ & + \beta_7 CATA_{i,t} + \sum_j \beta_{8,j} Dum_{i,t,j} + \beta_9 Gr_{Rate i,t} + \\ & \beta_{10} Inf_{i,t} + \beta_{11} Int_rate_{i,t} + K_i + \varepsilon_{i,t}. \end{aligned} \quad (2)$$

$$\begin{aligned} TR_{i,t} = & \alpha TR_{i,t-1} + \beta_1 SIZE_{i,t} + \beta_2 RAW_{INV i,t} + \beta_3 FIN_{INV i,t} \\ & + \beta_4 ROA_{i,t} + \beta_5 OTH_{LOAN i,t} + \beta_6 BK_{LOAN i,t} + \beta_7 CATA_{i,t} \\ & + \sum_j \beta_{8,j} Dum_{i,t,j} + \beta_9 Gr_{Rate i,t} + \beta_{10} Inf_{i,t} \\ & + \beta_{11} Int_rate_{i,t} + K_i + \varepsilon_{i,t}. \end{aligned} \quad (3)$$

As it is observed, the firm-specific and macro-indicators are same in Equations (2) and (3). The time dummy variables are selected as discussed later based on Bai and Perron (1998) statistics. The period prior to 2008 and 2013 is chosen as a reference category for Equations (2) and (3), respectively.

Results

The exploratory results are presented in the next subsection followed by the analytical findings.

Stylized Facts

The gross trade payables of the sample firms have marked continuous rise till 2015, although with decelerating growth rate, which has registered fall since afterwards (Figure 1). The gross trade payables dropped sharply by nearly 24% in 2018 compared to previous year. However, the average trade payable gives better insights about accounts payable as it normalizes using number of companies. It shows a steady rise in trade payable over the years with the growth rate of average accounts payable to be at 16% in 2015 to be the highest in recent times that has been less than 4% afterwards.

A similar trend is noted for the trade receivable as witnessed for the trade payable. Gross trade receivables have shown slowing growth rate before peaking in 2015 and declining thereafter (Figure 2). The average trade receivables have displayed a steady upsurge over the period, growing at a rate of 6% in 2016 and 2017 before a marginal fall in 2018. Both gross trade and average trade credit portrays Indian firms to be net receiver of the trade credit. Similar finding has been reported by Vaidya (2011) also for manufacturing firms in India.

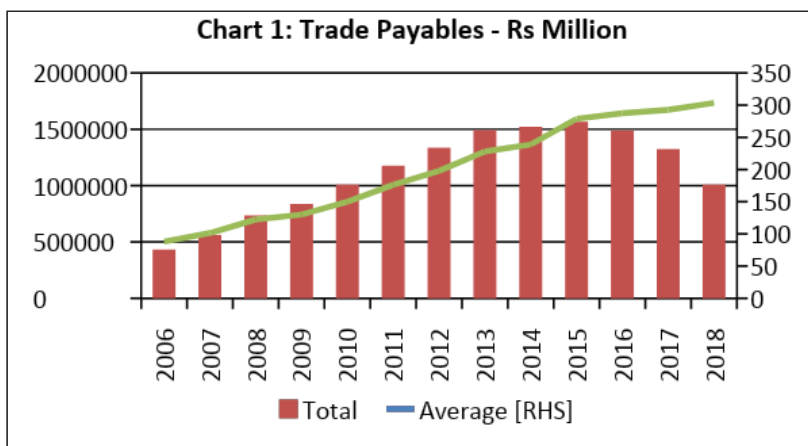


Figure 1. Trade Payables (₹ Million).

Source: The authors.

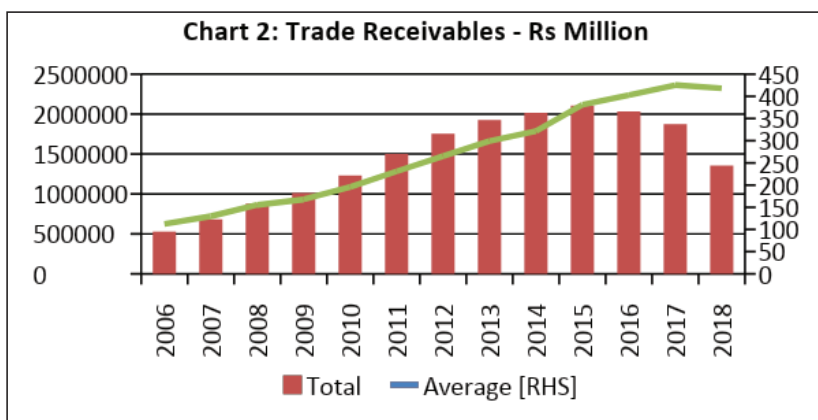


Figure 2. Trade Receivables (₹ Million).

Source: The authors.

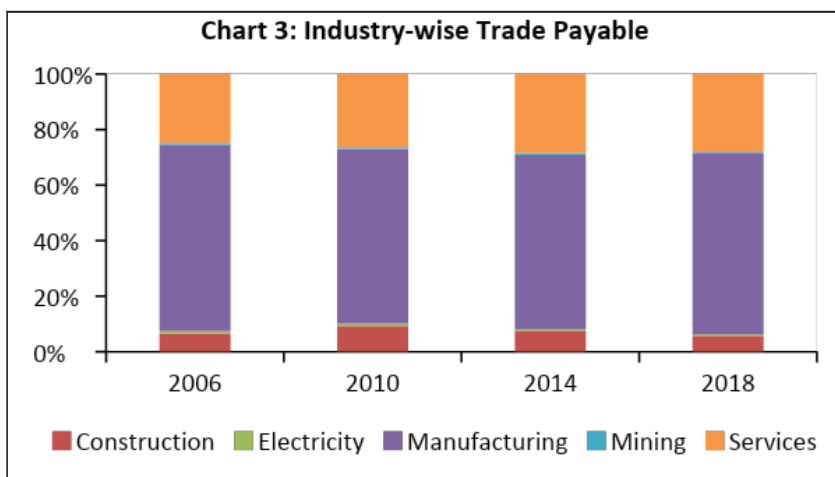


Figure 3. Industry-wise Trade Payable.

Source: The authors.

The industry-wise share of trade payables over the years depicts manufacturing industries occupying the majoring share of more than 62% across all the industries (Figure 3). It is followed by firms in service industry possessing around 25%–28% of overall trade payables. The construction sector holds less than 10% of industry share with a waning trend in recent times. In 2018, it contributed to 6% of the total trade payable share. Both electricity and mining are found to occupy the paltry portion of less than a percentage point for the entire study period.

The distribution of trade receivable is quite similar vis-a-vis trade payable again with manufacturing sector companies capturing the majority part with more than 62% for the selected time points (Figure 4). It is succeeded by service and construction sector comprising of 28% and 8% respectively in 2018.

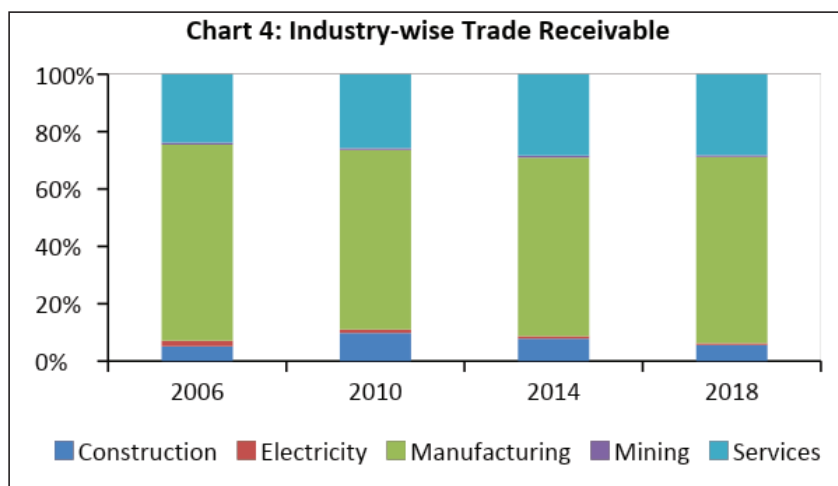


Figure 4. Industry-wise Trade Receivable.

Source: The authors.

Within the major industries, the trade credit of sub-industries is shown in Table 1. In respect of the trade payable for the mining sector, firms belonging to minerals have the highest trade payables that has retarded at 21% in 2018 compared to the previous year. The prominent categories within the manufacturing industry dealing in trade credit liabilities are chemicals and chemical products, transport equipment and machinery. Together firms belonging to these sub-groups comprise of nearly half of trade payable operations within manufacturing. Industrial construction firms form the significant portion of construction and real estate sector with more than 93% business in trade credit liability displayed a contraction of 37% in 2018 vis-a-vis 2017. Within electricity industry, firms associated with the electricity generation are the overwhelming majority as regards to trade payable that swelled from 83% to 96% from 2017 to 2018. Finally, wholesale and retail trading companies emerge as the predominant group in the service sector with more than half share in the trade credit liability that has shrunk by 31% in 2018 from 2017.

Table 1. Industry-wise Trade Credit.

(Amount in ₹ Million)					
Industry	Sub-industry	Trade Payables		Trade Receivables	
		2018	2017	2018	2017
Mining		5,139	8,211	8,439	12,124
		(0.51)	(0.62)	(0.62)	(0.65)
		289	2,081	146	2,541
	Coal and lignite	(0.03)	(0.16)	(0.01)	(0.14)

(Table 1 continued)

(Table 1 continued)

(Amount in ₹ Million)					
Industry	Sub-industry	Trade Payables		Trade Receivables	
		2018	2017	2018	2017
	Crude oil and natural gas	400 (0.04)	473 (0.04)	447 (0.03)	308 (0.02)
	Minerals	4,450 (0.44)	5,658 (0.43)	7,847 (0.58)	9,274 (0.49)
Manufacturing		658,296 (65.34)	827,733 (62.46)	880,697 (64.88)	1,146,241 (61.15)
	Chemicals and chemical products	122,587 (12.17)	145,682 (10.99)	188,298 (13.87)	229,497 (12.24)
	Construction materials	29,470 (2.92)	32,894 (2.48)	34,038 (2.51)	37,905 (2.02)
	Consumer goods	43,170 (4.28)	54,987 (4.15)	54,384 (4.01)	72,750 (3.88)
	Diversified	20,735 (2.06)	22,782 (1.72)	30,557 (2.25)	29,678 (1.58)
	Food and agro-based products	76,617 (7.6)	91,061 (6.87)	66,574 (4.9)	94,199 (5.03)
	Machinery	104,165 (10.34)	116,764 (8.81)	157,928 (11.63)	181,466 (9.68)
	Metals and metal products	74,170 (7.36)	128,418 (9.69)	102,218 (7.53)	179,217 (9.56)
	Miscellaneous manufacturing	14,020 (1.39)	19,808 (1.49)	21,901 (1.61)	36,447 (1.94)
	Textiles	75,031 (7.45)	85,628 (6.46)	120,352 (8.87)	147,023 (7.84)
	Transport equipment	98,330 (9.76)	129,709 (9.79)	104,447 (7.69)	138,060 (7.37)
Construction and real estate		57,997 (5.76)	93,419 (7.05)	76,973 (5.67)	138,164 (7.37)
	Industrial construction	54,828 (5.44)	87,259 (6.58)	73,022 (5.38)	130,094 (6.94)
	Real estate	3,169 (0.31)	6,159 (0.46)	3,951 (0.29)	8,071 (0.43)

(Table 1 continued)

(Table 1 continued)

(Amount in ₹ Million)					
Industry	Sub-industry	Trade Payables		Trade Receivables	
		2018	2017	2018	2017
Electricity		5,191	7,056	7,563	19,837
		(0.52)	(0.53)	(0.56)	(1.06)
		5,000	5,827	6,424	17,590
	Electricity generation	(0.5)	(0.44)	(0.47)	(0.94)
	Electricity transmission and distribution	191	1,229	1,139	2,247
		(0.02)	(0.09)	(0.08)	(0.12)
Services (other than financial)		280,922	388,877	383,721	558,062
		(27.88)	(29.34)	(28.27)	(29.77)
		3652	5456	7005	9983
	Communication services	(0.36)	(0.41)	(0.52)	(0.53)
		11,816	12,525	9,118	8,905
	Hotels and tourism	(1.17)	(0.95)	(0.67)	(0.48)
		16,548	19,853	44,142	54,078
	Information technology	(1.64)	(1.5)	(3.25)	(2.89)
		70,141	90,952	103,679	126,390
	Miscellaneous services	(6.96)	(6.86)	(7.64)	(6.74)
		21,884	34,087	34,968	58,713
	Transport services	(2.17)	(2.57)	(2.58)	(3.13)
		156,882	226,004	184,809	299,994
	Wholesale and retail trading	(15.57)	(17.05)	(13.62)	(16)
All		100,7544	1,325,296	1,357,393	1,874,428
		(100)	(100)	(100)	(100)

Source: Authors' calculation based on CMIE Prowess database.

Note: Figures in parenthesis are share in total

Turning to the trade credit liability side, units involved with minerals under the mining category are observed to be overwhelmingly delivering the trade credit to their partners' firms as a business strategy. In fact, within the mining classification the share of firms dealing with minerals has swelled from 77% to 93% from 2017 to 2018. In the manufacturing industry, the chemicals and chemical products group is the chief segment extending the trade credit with a share of more than 20% in 2017 and 2018. It is followed by machinery related companies whose share within the manufacturing sector has been in the range of 16%–18% in 2017 and 2018. In the construction and real estate group, it is the industrial and infrastructure conglomerates which comprise more than 94% share in the trade receivables.

Similarly, electricity generation emerges as the driving group of granting trade credit to its associates in the electricity industry. It captures a share of 85% in 2018 vis-a-vis 89% in 2017 marking a deceleration of -0.63% for the same duration. In the service sector, wholesalers and retailers record the highest share with 54% and 48% in 2017 and 2018, respectively. It is succeeded by miscellaneous services that have improved its share from 23% in 2017 to 27% in 2018.

Overall, the major industries dealing in the trade credit are manufacturing and services. Within these broad industry groups, few subgroups found to be substantially transacting both in utilizing and extending the trade credit seem to be wholesale and retail trading, chemical and chemical products, transport equipment, machinery, textiles and so on.

Firms have varied sources of external financing such as loans from bank, issuance of bonds, debentures and external commercial borrowings. Figure 5 displays the share of trade payables in contrast to other major instruments utilized by the private sector for their borrowing needs. Despite the requirement of certain formalities, bank lending remains as the main source of funding for corporate which has witnessed a decline by 36% in 2018 compared to a year ago. Trade payables emerge as the second highest funding option that has also recorded a fall by 24% for the same interval. Bank borrowing and trade payable together cover more than 80% of the financing need of the private sector. Other sources such as foreign currency borrowing, debentures and bonds and other sources have a share of 4%–6% each in total borrowings.

The mean account payable across major ownership categories shows the expansion of the average trade credit liability except the joint sector owned firms (Figure 6). Foreign-owned companies have the highest mean trade payables in 2018 that has grown at the rate of 10% in 2018 compared to 2017. The Indian private sector that encompasses the maximum number of firms in the sample has encountered an increase of 3% for the mean trade payable in 2018 vis-a-vis previous year.

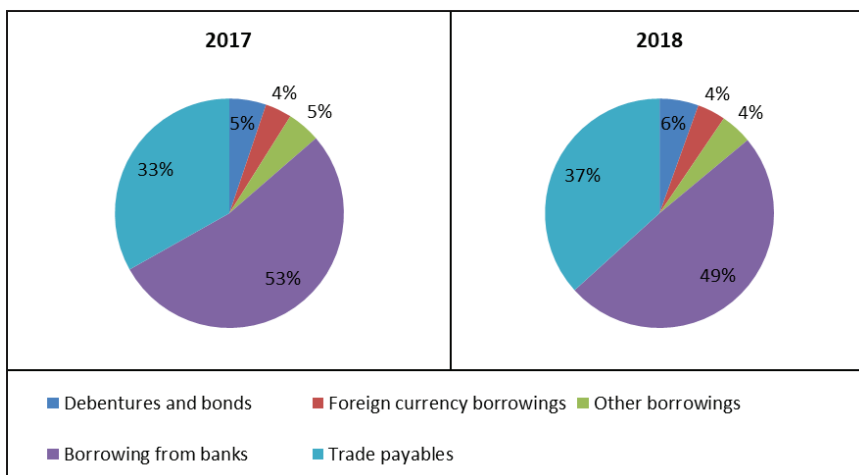


Figure 5. Trade Payables Vis-A-Vis Other Sources of Funds.

Source: The authors.

As noticed earlier for the trade payable, the average trade receivable is also the highest for the foreign private sector since last few years marking a growth rate of 15% in 2018 from a year ago (Figure 7). The joint sector has grown by 12%, however it has contracted slightly by 3% for Indian private businesses.

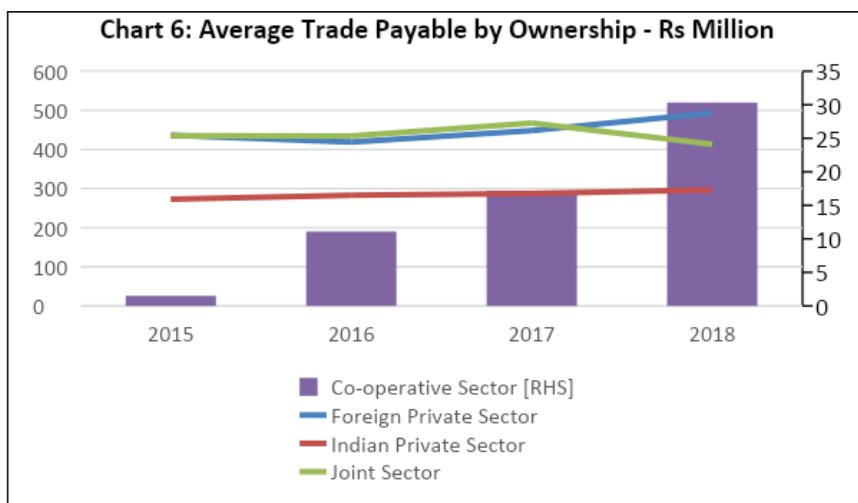


Figure 6. Average Trade Payable by Ownership (₹ Million).

Source: The authors.

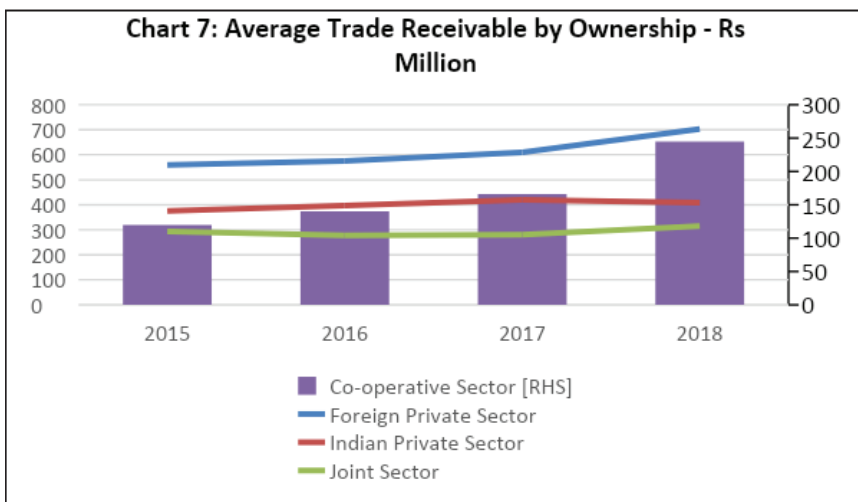


Figure 7. Average Trade Receivable by Ownership (₹ Million).

Source: The authors.

Longitudinal Analysis

Ahead of carrying out firm level analysis, a snapshot of micro-variables is shown in Table 2. The average trade payable is reported to be higher for smaller firms compared with larger units denoting more usage of the trade credit among smaller conglomerates albeit with higher variability. The trade receivable is higher for both bigger firms and at overall level in contrast with trade payable. The finding suggests that the Indian corporate sector is net receiver of the trade credit that owes the trade credit from its peers. Both the raw material and finished goods inventory indicate a slight decline in 2018 from 2017. The value of inventory is higher for smaller businesses than the larger companies. The average real sale of firms has slightly dipped in 2018 from the earlier year. Both the sales and profitability of small-sized companies is lower vis-a-vis bigger concerns. It is observed that both bank and other borrowing of a representative company have risen in 2018 from its earlier level. The quantum and rise of both borrowings is sharper for smaller firms relative to their larger counterparts. For smaller businesses the annual hike is 43% and 27% respectively for bank and other borrowings respectively in 2018. CATA has marked a slight decline in 2018 from 2017. However, its level is quite similar across various sized conglomerates.

Table 2. Average of Variables by Firm Size.

Year	2018			2017		
Variables	Small	Large	Overall	Small	Large	Overall
Trade payable to asset	29.43 (192.18)	13.19 (13.87)	17.82 (103.43)	28.67 (177.56)	12.75 (13.45)	16.88 (91.41)
Trade receivable to asset	25.52 (24.66)	18.07 (15.85)	20.23 (19.13)	24.93 (24.7)	18.35 (16.23)	20.05 (19.01)
Raw material inventory to asset	8.99 (10.02)	6.96 (6.53)	7.22 (7.09)	9.11 (11.86)	7.43 (7.21)	7.64 (7.95)
Finished goods inventory to asset	13.8 (16.07)	10.25 (10.7)	10.92 (11.98)	16.7 (19.81)	10.71 (11.15)	11.82 (13.38)
Log (real sales)	2.24 (1.95)	7.58 (1.36)	5.94 (2.92)	2.39 (2.04)	7.64 (1.34)	6.18 (2.82)
ROA	-1.15 (5.69)	2.17 (6.39)	0.91 (6.34)	-0.92 (5.54)	2.19 (6.1)	1.08 (6.09)
Bank borrowing to asset	62.88 (362.52)	22.92 (20.15)	28.2 (133.37)	43.89 (289.33)	25.1 (19.86)	27.61 (107.38)
Other borrowing to asset	70.97 (395.64)	11.28 (17.19)	19.16 (145.65)	55.71 (320.38)	11.58 (16.23)	17.48 (118.84)

(Table 2 continued)

(Table 2 continued)

Year	2018			2017		
Variables	Small	Large	Overall	Small	Large	Overall
Current asset to total asset	39.87 (32.6)	39.16 (24.68)	39.41 (27.82)	41.55 (33.81)	40.45 (24.76)	40.82 (28.16)

Source: The authors.

Notes: Firms are classified as 'Small' that have assets less than 25 percentile score. 'Large' firms are those having assets percentile score of more than 75. Figures in bracket denote the standard deviation.

ROA: return on assets.

Both parametric and non-parametric statistical tests have been performed to check if the conduct of the trade credit is different across various industries and firm size. The results signify that the quanta of trade credit both across industries and size does significantly vary (Table 3). As our dataset is reasonably wide, we also perform the Bai and Perron (1998) test to identify structural breaks during the period of analysis possibly due to financial crisis and revisions in the industrial policy. For accounts payable the statistics suggest three break periods as 2007, 2011 and 2015. Whereas for accounts receivable the break years correspond to 2012 and 2015. The year 2007 is just prior to the financial crisis. Break in 2011 seems to be due to the European sovereign debt crisis in 2011–2012 which led to the decline in demand from Euro area and affected industrial production in India. Besides, the economy had the high fiscal deficit of nearly 7% of GDP, elevated inflation rates during 2014–2015 that could have influence on the corporate activity.

The dynamic panel regression model applying GMM estimator with accounts payables as dependent variable is reported in Table 4. Separate regressions were performed for all firms, major industries, namely manufacturing and services. Firms existing in the top/bottom 25 percentile as per total assets are chosen for small/large classification and estimation is performed to independently identify and analyze the varying impact of various causal factors. Most of the estimates are found to be significant signifying satisfactory fit. Initiating with Column (1), estimation results for the entire sample show positive and significant lag coefficient

Table 3. Tests of Equality.

Test	Industry		Firm Size	
	Trade Payable	Trade Receivable	Trade Payable	Trade Receivable
ANOVA	11	132	103	464
Kruskal–Wallis Statistics	3,058	1,260	24	58
Van der Waerden	2,802	1,144	120	172

Source: The authors.

Note: All values significant at 1% significance level.

ANOVA, analysis of variance.

Table 4. Trade Payable.

Variables	Overall	Manufacturing	Services	Small	Large
	(1)	(2)	(3)	(4)	(5)
	0.298***	0.279**	0.164***	0.193**	0.578***
LI.Dep	(0.106)	(0.129)	(0.047)	(0.09)	(0.085)
	1.239***	1.277***	2.027***	1.898***	0.695***
ln(Sales Real)	(0.155)	(0.172)	(0.348)	(0.49)	(0.218)
Inventory (Raw Material) to Asset	0.049***	0.056***	-0.013	-0.002	0.083***
	(0.012)	(0.012)	(0.049)	(0.04)	(0.023)
Inventory (Finished Goods) to Asset	-0.035***	-0.02*	-0.055*	-0.05	0.023
	(0.011)	(0.012)	(0.031)	(0.041)	(0.019)
	-0.146***	-0.145***	-0.166***	-0.118***	-0.176***
ROA	(0.012)	(0.013)	(0.041)	(0.043)	(0.019)
Other Borrowing to Asset	-0.137***	-0.148***	-0.139***	-0.072***	-0.161***
	(0.007)	(0.008)	(0.016)	(0.012)	(0.01)
Bank Borrowing to Asset	-0.162***	-0.155***	-0.308***	-0.053***	-0.247***
	(0.007)	(0.008)	(0.019)	(0.009)	(0.011)
Current Asset to Total Asset	0.224***	0.204***	0.349***	0.234***	0.185***
	(0.007)	(0.008)	(0.023)	(0.029)	(0.012)
	-0.898***	-0.995***	-0.094	-0.653	0.278
Year Dum2	(0.195)	(0.2)	(0.776)	(0.88)	(0.321)
	-1.644***	-1.778***	-0.859	-1.181	0.363
Year Dum3	(0.298)	(0.314)	(1.106)	(1.314)	(0.487)
	-1.718***	-1.791***	-0.821	-0.405	0.085
Year Dum4	(0.302)	(0.33)	(1.055)	(1.295)	(0.499)
	0.041	0.032	0.031	-0.124	-0.099*
LI.Gr_Rate	(0.038)	(0.044)	(0.121)	(0.14)	(0.054)
	0.054	0.08**	-0.084	0.078	-0.054
LI.Inf	(0.033)	(0.033)	(0.128)	(0.151)	(0.051)
	0.279***	0.268***	0.294	-0.104	0.044
LI.Int_rate	(0.053)	(0.057)	(0.201)	(0.242)	(0.082)
<i>Model diagnostics</i>					
AR-1	-4.13***	-3.28***	-6.51***	-4.58***	-7.53***
AR-2	-1.56	-1.19	-3.01***	0.70	1.16
Sargan	0.13	0.15	0.13	0.32	1.93

Source: Authors' calculation.

Notes: *, **, *** indicate significance at 10%, 5% and 1% level, respectively.
ROA, return on assets.

reflecting the strong persistence. The raw material inventory has positive and significant effect on the accounts payable. It indicates higher usage of trade payable towards accumulating inventories. The finding is in consonance with Vaidya (2011) that also found positive relation for raw material inventory with accounts payables for Indian firms operating during 1996–2006. However, an indirect connection is emanating between trade payable and finished goods inventory. The outcome suggests that the account payable is chiefly utilized for the procurement of raw material. SIZE is having a positive and statistically significant impact on the trade payable. It shows that bigger firms earning higher revenue receive more trade credit as they enjoy greater trustworthiness and market confidence (Casey & O'Toole, 2014). Bougheas et al. (2009) also found similar finding for the manufacturing sector companies in the UK although Vaidya (2011) found it significant only for account receivable. ROA is having strong inverse impact on accounts payables implying profitable firms preferring retained earnings compared to the trade credit that is more expensive form of the credit. The result is similar as also found by Vaidya (2011) but contrary to that of Bougheas et al. (2009). The coefficient of OTH_LOAN and BK_LOAN is negative and significant. A higher financing from other sources and banks is leading to the reduced trade credit borrowing indicating that both other financing sources and banks are used as substitute vis-a-vis trade credit. All, ROA, OTH_LOAN and BK_LOAN are in harmony with the pecking order theory that postulates internally generated funds to be higher in order compared to more expensive forms of credit (Myers & Majluf, 1984). Liquidity of firm as captured by current ratio is significant and positive. Higher liquidity affords the higher trade credit liability to a firm. The finding is consistent as per Bougheas et al. (2009) and Vaidya (2011). The year dummy variables are strongly negative implying decline in the trade credit since the pre-crisis period. Among macro indicators, the higher interest rate is having positive influence on account payables. Higher rates narrow down the rate gap between formal sources of the credit and the trade credit leading to the greater usage of the trade credit that is relatively convenient source of finance. Continuing with other columns of Table 4, it is found that most of the results obtained at the aggregate level for all the firms hold for other classifications also. SIZE is consistently positive and statistically significant for other categories of enterprises also. The outcome implies that larger firms are receiving more trade credit due to their creditworthiness and reputational considerations with potential buyers (Petersen & Rajan, 1997). The raw material inventory is strongly positive for manufacturing and large companies, however turning insignificant for firms involved in services and small groupings. This may be observed due to the low usage of raw materials and inaccessibility of the trade credit for services and small categories of conglomerates, respectively. The service sector such as software firms, trade, communications and transportation predominantly comprises intangible goods, where the role of the raw material inventory is limited. Profitability, OTH_LOAN and BK_LOAN are steadily sharing an inverse association, whereas it is the opposite connection for CATA across all segments. The year dummies are turning to be significant only for the manufacturing cluster. Moving to economic parameters, inflation and interest rate

have a positive impact on the trade payable for the manufacturing sector. A negative coefficient of the GDP growth rate for large firms is evidenced that indicates counter-cyclicality.

Moving forward, GMM estimation results with accounts' receivables as dependent variable are reported in Table 5. As earlier, Columns (1)–(5) denote full sample, manufacturing, services, small and large units, respectively. A significant positive estimate for lagged dependent is evident for all firm, that is, Column (1) like trade payable implying strong persistence for trade receivable also. A direct positive impact of SIZE on account receivables is distinctly visible. The finding confirms that bigger firms are also the biggest lenders of the trade credit in line with Petersen and Rajan (1997), Bougheas et al. (2009) and Vaidya (2011). The size of both raw material and finished goods inventory are having strong negative effect on accounts receivable. The outcomes suggest that firms having reasonable stock of raw or finished goods have less incentive to offer credit for obtaining additional raw stock or selling their final goods respectively leading to inverse relationship that is also supported by Bougheas et al. (2009) and Vaidya (2011). Profitability as captured by ROA is having significant positive influence on the

Table 5. Trade Receivable.

Variables	Overall	Manufacturing	Services	Small	Large
	(1)	(2)	(3)	(4)	(5)
	0.231***	0.25***	0.091*	0.579**	0.084**
L1.Dep	(0.067)	(0.071)	(0.054)	(0.226)	(0.035)
ln(Sales	1.362***	1.413***	0.846***	0.93**	1.159***
Real)	(0.09)	(0.102)	(0.196)	(0.434)	(0.128)
Inventory	−0.703***	−0.69***	−0.849***	−0.818***	−0.566***
(Raw Mate-	(0.014)	(0.015)	(0.03)	(0.052)	(0.014)
rial) to Asset					
Inventory	−0.721***	−0.71***	−0.81***	−0.913***	−0.55***
(Finished	(0.014)	(0.015)	(0.02)	(0.072)	(0.012)
Goods) to					
Asset					
	0.024***	0.018**	0.005	0.016	0.046***
ROA	(0.008)	(0.009)	(0.023)	(0.038)	(0.011)
Other Bor-	0.004	−0.004	−0.018**	0.004	0.035***
rowing to	(0.003)	(0.004)	(0.009)	(0.01)	(0.005)
Asset					
Bank Bor-	0.007***	0.005	−0.014	−0.002	0.031***
rowing to	(0.003)	(0.003)	(0.011)	(0.007)	(0.005)
Asset					
Current As-	0.643***	0.634***	0.783***	0.789***	0.516***
set to Total	(0.008)	(0.009)	(0.014)	(0.039)	(0.007)
Asset					

(Table 5 continued)

(Table 5 continued)

Variables	Overall	Manufacturing	Services	Small	Large
	(1)	(2)	(3)	(4)	(5)
	0.094	0.134	-0.224	-0.444	0.406***
Year Dum2	(0.099)	(0.104)	(0.331)	(0.683)	(0.14)
	0.464***	0.535***	-0.136	-1.15	0.833***
Year Dum3	(0.134)	(0.141)	(0.436)	(0.992)	(0.178)
	-0.018	-0.032	0.041	-0.139	0.016
LI.Gr_Rate	(0.02)	(0.021)	(0.06)	(0.103)	(0.026)
	-0.017	-0.017	-0.042	-0.124	0.019
LI.Inf	(0.017)	(0.017)	(0.052)	(0.089)	(0.023)
	0.019	-3.0E-04	0.101	-0.247	0.011
LI.Int_rate	(0.026)	(0.028)	(0.082)	(0.154)	(0.036)
Model Diagnostics					
AR-1	-6.79***	-6.81***	-5.81***	-3.44***	-8.12***
AR-2	-0.66	-0.06	-1.31	-0.32	-3.93***
Sargan	0.44	0.81	0.01	0.12	1.28

Source: Authors' calculation.

Notes: *, **, *** indicate significance at 10%, 5% and 1% level, respectively.

ROA, return on assets.

trade credit receivable. It represents net earnings to be used as a tool towards extending more credit. The finding is contrary to Vaidya (2011) that found significant negative impact of net profits. However, the result is consistent as obtained by Petersen and Rajan (1997) and Bougheas et al. (2009). The other sources of finance are turning to be positive but insignificant. But Bank borrowings' coefficient is positive and significant implying firms borrowing more are also extending more trade credit. It points that bank borrowing and account receivables are complement to each other for the overall scenario. Generally, other studies such as Petersen and Rajan (1997), Bougheas et al. (2009) and Ghosh (2015) also found similar result although Vaidya (2011) found an inverse relationship between bank borrowing and accounts receivable. Liquidity (CATA) has a strong positive impact on account receivable. It corroborates the usage of additional financial resources towards extending credit to potential clients, that is, in tandem with Vaidya (2011). The positive estimate for year dummies indicate the higher usage in the respective period following financial crisis. The coefficient of the macro-economic growth rate and inflation are having negative sign albeit statistically insignificant. Similarly, the interest rate is also derived to be insignificant although positively associated with trade receivable. The outcomes suggest the limited influence of macro climate for trade receivable. Proceeding with other company classifications, that is, Columns (2)–(5), it is found that qualitatively, the results are similar for most of the parameters. OTH_LOAN has a negative estimate for services and positive for larger units.

Next, the basic specification tests have been carried out to ascertain the consistency of GMM estimation for both trade payable and trade receivable. The AR-2, that is, the second-order serial correlation of the difference GMM regression is insignificant in most of the cases implying rejection of the serial correlation. The other important test is the Sargan statistics of over-identifying restrictions, which examines the validity of instruments. It establishes the authenticity of the instruments and hence the consistency of estimates.

Prominent outcomes include persistency in trade credit transactions, predominant role of inventory management decisions for both usages of the trade credit and its disbursement. Bigger firms are indulging in both higher usage and disbursement of the trade credit. The pecking order theory is well supported with firms preferring net earnings over more expensive form of the trade credit for financing requirements. Bank borrowing is found to complement trade credit receivables corroborating findings of Bougheas et al. (2009) and Ghosh (2015). Among macro indicators, higher interest rates incentivize companies to move in favour of accounts payables, whereas high inflation rate enlarges the trade payable liability for the manufacturing sector.

The primary policy input emanating from the study is that of the requirement of the structured institutional arrangement for financial needs of businesses. Otherwise, lack of accessibility of bank credit forces companies to depend on more expensive form of financing, that is, trade credit. Hence, more awareness and convenient structure is desired from formal financial institutions. Also, robust economic climate with low interest and inflation rate regime facilitates the sensible trade credit culture.

Conclusion

The trade credit is a vital financial management medium among businesses used with various objectives. The study focuses on the determinants of the trade credit for the Indian corporate sector. It not only includes the firm-level characteristics but also macro factors to understand the relation of the trade credit with firm level micro factors and economic cycles. Employing a rich dataset covering the crisis period operating in multiple sectors such as manufacturing, services, construction and others, it identifies the structural breaks before performing the analysis. Apart from sharing long-term relationship with regards to trade credit, nuances of the inventory management are established. While the trade payable is mainly driven by the raw material inventory, firms having reasonable stock of raw or finished goods inventory are less likely to offer the trade credit. Bigger firms are both high users and the supplier of the trade credit. The financing from formal sources is acting as a substitute to the trade credit usage. It clearly indicates that limited access to specialized financing institutions forces firms to buy raw material on trade credit, which is in fact more expensive. Strong and accessible banking and non-banking institutions can help in alleviating this bottleneck. Last but not the least, the economic climate does play a role in the trade credit operations especially

for trade credit borrowings. While the manufacturing sector is affected adversely with rising interest and inflation rates, larger companies exhibit counter-cyclicalities, which undoubtedly points to importance of stable economic conditions for business prosperity.

Acknowledgement

The authors are thankful for the useful comments provided by the reviewer(s) on an earlier draft. However, the views expressed in the study are purely personal opinion of the authors.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship and/or publication of this article.

Notes

1. GDP deflator has been utilized to derive real sales.
2. Other profitability measures like profit after tax as percentage of total income, return on net worth were also employed but were dropped from final results due to insignificance.
3. Annual average figures are employed.

References

- Arellano, M., & Bond, S. (1991). Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations. *The Review of Economic Studies*, 58(2), 277–297.
- Arellano, M., & Bover, O. (1995). Another look at the instrumental variable estimation of error-components models. *Journal of Econometrics*, 68(1), 29–51.
- Bai, J., & Perron, P. (1998). Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1), 47–78.
- Biais, B., & Gollier, C. (1997). Trade credit and credit rationing. *The Review of Financial Studies*, 10(4), 903–937.
- Bougheas, S., Mateut, S., & Mizen, P. (2009). Corporate trade credit and inventories: New evidence of a trade-off from accounts payable and receivable. *Journal of Banking & Finance*, 33(2), 300–307.
- Burkart, M., & Ellingsen, T. (2004). In-kind finance: A theory of trade credit. *American Economic Review*, 94(3), 569–590.
- Casey, E., & O'Toole, C. M. (2014). Bank lending constraints, trade credit and alternative financing during the financial crisis: Evidence from European SMEs. *Journal of Corporate Finance*, 27, 173–193.
- Coulibaly, B., Sapriza, H., & Zlate, A. (2013). Financial frictions, trade credit, and the 2008–09 global financial crisis. *International Review of Economics & Finance*, 26, 25–38.

- Cunat, V. (2007). Trade credit: Suppliers as debt collectors and insurance providers. *The Review of Financial Studies*, 20(2), 491–527.
- Du, J., Tao, Z., & Lu, Y. (2008). *Bank loans and trade credit under China's financial repression*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1495600
- Ghosh, S. (2015). Trade credit, bank credit and crisis: Some empirical evidence for India? *Margin: The Journal of Applied Economic Research*, 9(4), 333–361.
- Klapper, L., Laeven, L., & Rajan, R. (2011). Trade credit contracts. *The Review of Financial Studies*, 25(3), 838–867.
- Kumar, P., & Shrivastava, A. (2013). Trade credits and corporate profitability: Some empirical findings in Indian context. *Journal of Management Value & Ethics*, 3(3), 41–57.
- Love, I., & Zaidi, R. (2010). Trade credit, bank credit and financial crisis. *International Review of Finance*, 10(1), 125–147.
- Molina, C. A., & Preve, L. A. (2009). Trade receivables policy of distressed firms and its effect on the costs of financial distress. *Financial Management*, 38(3), 663–686.
- Myers, S. C., & Majluf, N. S. (1984). Corporate financing and investment decisions when firms have information that investors do not have. *Journal of Financial Economics*, 13(2), 187–221.
- Petersen, M. A., & Rajan, R. G. (1997). Trade credit: Theories and evidence. *The Review of Financial Studies*, 10(3), 661–691.
- Smith, J. K. (1987). Trade credit and informational asymmetry. *The Journal of Finance*, 42(4), 863–872.
- Vaidya, R. R. (2011). The determinants of trade credit: Evidence from Indian manufacturing firms. *Modern Economy*, 2(05), 707.
- Wilner, B. S. (2000). The exploitation of relationships in financial distress: The case of trade credit. *The Journal of Finance*, 55(1), 153–178.
- Yang, X. (2011). Trade credit versus bank credit: Evidence from corporate inventory financing. *The Quarterly Review of Economics and Finance*, 51(4), 419–434.